MIL-G-50440 (MU)
15 August 1969
SUPERSEDING
(See 6.8)

MILITARY SPECIFICATION

GRENADE, GP: M43A1,
PARTS FOR

1. SCOPE

1.1 This specification covers the parts for one type of grenade designated as Grenade, GP: M43A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Box, Fiberboard

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing, Interior Packaging
MIL-B-2427	-	Boxes, Ammunition Packing; Wood, Nailed
MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

FSC: 1330

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PURCHASE SPECIFICATIONS

AEI-93	-	Primer, M104, LAP

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

US ARMY MUNITIONS COMMAND

DP-119549	-	Grenade, GP, M43A1
DP-119552	-	Housing Assembly
C8796522	-	Marking Diagram and Sealing for Wood Packing Boxes

PUBLICATIONS

ARMY

ET-119552	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify 49 CFR 171-179 (latest revision).)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-64	-	Standard Method of Salt Spray (Fog) Testing
ASTM E 8-66	-	Tension Testing of Metallic Materials
ASTM E 18-67	-	Standard Methods of Test for Rockwell Hardness & Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The housing assembly shall comply with all requirements specified on Drawing (Dwg.) P-119552, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Security of hinge pins and outer springs.-The hinge pins and outer springs in housing assembly shall withstand a force imparted by a free-falling weight of eight (8) pounds, minimum (min), dropped from a height of five (5) inches, min, in a direction perpendicular to the long axis of the outer spring and at 1.50 plus or minus (±) .06 inch distance from the centerline of the hinge pin without shearing the hinge pin or cracking of the outer spring (hinge pins may deform, but must remain affixed to the housing) when tested as specified in 4.4.19.

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3.4 Simulated grenade catapult test (when applicable).-The simulated grenade shall exhibit an average impact velocity of one-hundred and twenty-five (125) plus or minus (±) ten (10) feet per second (fps) and shall evidence no separation of the nose housing when tested as specified in 4.4.21.

3.5 Workmanship.-All parts and assemblies shall be fabricated, finished, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, unblended radii, cracks, surface defects, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

3.6 Initial qualifying test for housing assembly pack.-The fiber box or wooden box, with housing assemblies, shall be subjected to the drop test specified in 4.4.25 and shall show no evidence of spilled or shifted contents. In addition, the safety clip shall remain securely attached to the housing assembly, shall not have moved from its position as specified on the housing assembly drawing, and the housing assembly shall show no change or damage.

3.7 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (see 6.7).

3.8 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract (see 6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

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4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Ten (10) nose housings (after heat treatment and prior to machining).
b. One-hundred and ninety-five (195) nose housings (Dwg. FP-119548).
c. Seventy-five (75) of each part.
d. Seventy-five (75) of each subassembly.
e. Seventy-five (75) of each assembly.
f. One-hundred and fifty (150) housing assemblies (Dwg. DP-119552).
g. In rod form, ten rods of Lower Pin bar stock, or if in coil form, six samples from each coil (See Dwg. BP-119560). Samples shall be selected after final heat treatment and forming processes but do not have to be finish-machined.

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All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. After classification of defects and other inspections have been performed as specified in 4.3.2, the following quantities shall be subjected to the indicated tests.

4.2.2.1 Mechanical properties of nose housing (after heat treatment and prior to machining (See Dwg. FP-119548).-One (1) tension test specimen shall be taken from each of the ten (10) nose housings in accordance with the following and tested as specified in 4.4.1.

a. Forgings - as indicated in Figure 1.
b. Bar stock - as indicated in ASTM E8-66, Figure 9, Specimen 1. The specimen shall be taken from the center of the area of the largest possible inscribed circle on the face of the nose housing as noted on the applicable drawing.

4.2.2.2 Hardness of hinge pin and outer spring (See Dwg. CXP-113760 and FP-119554, resp.).-The seventy-five (75) hinge pins and outer springs shall be subjected to this test as specified in 4.4.2.

4.2.2.3 Deflection-load test of leaf springs of outer spring (See Dwg. FP-119554).-Twenty-five (25) outer springs shall be subjected to this test as specified in 4.4.3.

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4.2.2.4 Spring test of side tabs of outer spring (See Dwg. FP-119554).-Following completion of the test specified in 4.2.2.3, the twenty-five (25) outer springs shall be subjected to this test as specified in 4.4.4.

4.2.2.5 Leakage test of diaphragm (See Dwg. CXP-113757).-The seventy-five (75) diaphragms shall be subjected to this test as specified in 4.4.5.

4.2.2.6 Deformation test of nose housing (See Dwg. FP-119548).-Twenty-five (25) nose housings shall be subjected to this test as specified in 4.4.6.

4.2.2.7 Direction of grain.

a. Main yoke spring (See Dwg. CP-119558).
b. Secondary yoke spring (See Dwg. CP-119559).
c. Safety clip (See Dwg. DXP-113789).

Twenty-five (25) of each part specified in a, b, and c shall be subjected to this test as specified in 4.4.7.

4.2.2.8 Pull load test of safety clip (See Dwg. DXP-113789).-Twenty-five (25) safety clips shall be subjected to this test as specified in 4.4.8.

4.2.2.9 Load test of yoke (See Dwg. DP-119557).-Twenty-five (25) yokes shall be subjected to this test as specified in 4.4.9.

4.2.2.10 Load tests of slide spring (after temperature conditioning)(See Dwg. CXP-113763).-The seventy-five (75) slide springs shall be subjected to these tests as specified in 4.4.10.

4.2.2.11 Salt spray test of hinge pins (See Dwg. CXP-113760).-Twenty-five (25) hinge pins shall be subjected to this test as specified in 4.4.11.

4.2.2.12 Shear strength test of lower pin rod or coil stock (See Dwg. BP-119560).-In rod form, two (2) specimens shall be taken from each of the ten (10) rods. The specimens shall consist of one (1) taken from the middle and one (1) taken from either end. In coil form, two (2) specimens shall be taken from each end of the coil and two (2) specimens taken at the approximate center of each coil, with a six (6) inch minimum distance between specimens. The test shall be performed as specified in 4.4.12.

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4.2.2.13 Push test of yoke springs and lower pin of yoke spring assembly (See Dwg. DP-119556).-The seventy-five (75) yoke spring assemblies shall be subjected to this test as specified in 4.4.13.

4.2.2.14 Deflection-load test of yoke springs of yoke spring assembly (See Dwg. DP-119556).-The seventy-five (75) yoke spring assemblies shall be subjected to this test as specified in 4.4.14

4.2.2.15 Push-out test of restricter of plug-restricter assembly (See Dwg. BXP-113764).-The seventy-five (75) plug-restricter assemblies shall be subjected to this test as specified in 4.4.15.

4.2.2.16 Air Flow test of plug-restricter assembly (See Dwg. BXP-113764).-The seventy-five (75) plug-restricter assemblies shall be subjected to this test as specified in 4.4.16.

CAUTION: The housing assembly in the following tests contains an explosive component.

4.2.2.17 Push test of lower sleeve in housing assembly (see Dwg. DP-119552).-After completion of test specified in 4.2.4.1, the lower sleeves of the seventy-five (75) housing assemblies shall be subjected to this test as specified in 4.4.24.

4.2.2.18 Security of hinge pins and outersprings (See 3.3).-After completion of test specified in 4.2.4.1, the hinge pins and outer springs of the seventy-five (75) housing assemblies shall be subjected to this test as specified in 4.4.19.

4.2.2.19 Push-out test of plug-restricter assembly (See Dwg. DP-119552).-After completion of test specified in 4.2.4.1, the plug-restricters of the seventy-five (75) housing assemblies shall be subjected to this test as specified in 4.4.23.

4.2.2.20 Freeness of pivoting of outer springs (See Dwg. DP-119552).-The seventy-five (75) housing assemblies shall be subjected to the test as specified in 4.4.18.

4.2.2.21 Push test of primer of slide assembly (See Dwg. CP-119555).-After completion of test specified in 4.2.4.1, the seventy-five (75) slide assemblies shall be subjected to this test as specified in 4.4.17.

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4.2.3 Rejection.-If any specimen, part, or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a specimen, part or assembly in the sample to comply with any of stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Additional first article inspection.

4.2.4.1 Delay arming time of housing assembly (See Dwg. DP-119552).-The seventy-five (75) housing assemblies shall be subjected to this test as specified in 4.4.22.

4.2.4.1.1 Rejection.-If three (3) or more housing assemblies exceed the maximum allowable delay time specified on the applicable drawing, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities, as necessary, until such time as a quantity successfully passes the inspections specified, or until otherwise directed by the procuring activity.

4.2.4.2 Nose housing qualification test (See Dwg. FP-119548).-One-hundred and twenty (120) nose housings (Dwg. FP-119548) shall be subjected to this test as specified in 4.4.20.

4.2.4.2.1 Rejection.-The test data shall be evaluated in accordance with applicable special instructions and, if value obtained, does not exceed three and one-half ($3\frac{1}{2}$) inches, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities, as necessary, until such time as a quantity successfully passes the inspections specified, or until otherwise directed by the procuring activity.

4.2.4.3 Simulated grenade catapult test (See 3.4).-Seventy-five (75) housing assemblies (See 6.5) shall be assembled with a Government-furnished container assembly, and a ball to simulate the weight and size as specified on Dwg. 8865585). Assembly shall be in accordance with Dwg. P-119549. Instrumentation shall be provided to check impact velocity on every fifth sample shot. Velocity results shall be recorded (See 6.6). The test shall be performed as specified in 4.4.21.

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4.2.4.3.1 Results to be discarded.-Discard results for the following and provide additional grenades to complete the sample quantity specified:

a. Impact velocity less than 100 feet per second (fps)
b. Equipment malfunction.
c. Poor orientation on impact.

4.2.4.3.2 Records.-Record for information, the following:

a. Impact velocity or nose housing malfunctions.
b. Crushed or cracked nose housing.

4.2.4.3.3 Rejection.-If one or more grenades fail to comply with the specified requirements, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity as specified by the contracting officer.

4.2.4.4 Initial qualifying tests for housing assembly pack (See 3.6).-Five fiber boxes or wooden boxes, with housing assemblies, shall be subjected to the drop tests as specified in 4.4.25. If any pack fails to comply with the specified requirements, the first article quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new first article quantity as specified by the Contracting Officer.

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. In each inspection lot of grenades only the

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following components and assemblies shall be required to be from one (1) lot interfix number from one manufacturer:

a. Nose housings
b. Outer springs
c. Primers (See 4.3.3.3)

NOTE: For nose housings, one lot interfix number from one manufacturer requires the lot interfix number to be related to the aluminum alloy 7001 material that is produced by one manufacturer, in one unchanged process, etc.

4.3.1.1 Heat treatment lot (nose housings).-A heat treatment lot shall consist of nose housings (forgings or bar-stock) of one lot of 7001 aluminum alloy that has been heat treated by the same process, in the same furnace, at the same temperature for the same length of time. Identity shall be maintained until all mechanical properties tests have been completed. (See 4.3,3.5).

4.3.1.1.1 A heat treatment lot of hose housings (forgings) shall not exceed 35,000 pieces.

4.3.1.1.2 A heat treatment lot of bar-stock shall not exceed 11,000 pounds of bar-stock.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. In addition, at the option of the procuring activity, defect characteristics of the same classification which have been separated into two (2) groups to specifically allow the optional method of inspection in accordance with 4.3.3.1, 4.3.3.2 and 4.3.3.4 may be combined into one (1) group when the option of 4.3.3.1, 4.3.3.2 and 4.3.3.4. is not exercised. The AQL for the combined grouping will be as indicated in the applicable classification of defects paragraph. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Diaphragm (See Dwg. CXP-113757 covering a detail of Dwg. DP-119552).-

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	None defined		
Minor	AQL 0.65 Percent (See Para 4.3.3.1)		
201.	Evidence of poor workmanship (See 3.5)	Visual	01001

4.3.2.2 Sleeve, lower (See Dwg. BXP-113761 covering a detail of Dwg. DP-119552).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Height	Gage	02001
102.	Inside diameter	Gage	02002
103.	Outside diameter, min	Gage	02003
104.	Concentricity of inside diameter with outside diameter	Gage	02004
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	02005

4.3.2.3 Slide (See Dwg. DP-119561 covering a detail of Dwg. CP-119555).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.50 Percent (See Para 4.3.3.1)		
101.	Largest outside diameter (round end)	Gage	03001
102.	Concentricity of second largest outside (flat end) diameter with largest outside diameter	Gage	03002

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103.	Concentricity of primer flash hole with primer cavity	Gage	03003
104.	Total length, max	Gage	03004
105.	Diameter of lower pin hole	Gage	03005
106.	Location of primer cavity	Gage	03006
107.	Location of lower pin hole	Gage	03007
108.	Length from top to bottom flat	Gage	03008
109.	Location of shoulder, largest outside diameter	Gage	03009
110.	Length to second largest outside diameter, min	Gage	03010
111.	Depth of primer cavity, max	Gage	03011
112.	Second largest outside diameter	Gage	03012
113.	Incomplete casting	Visual	03013
Minor	AQL 2.50 Percent (See Para 4.3.3.1)		
201.	Third largest outside diameter, max	Gage	03014
202.	Concentricity of third largest outside diameter with largest outside diameter	Gage	03015
203.	Diameter of primer flash hole	Gage	03016
204.	Width across flat, min, largest outside diameter	Gage	03017
205.	Evidence of poor workmanship (See 3.5)	Visual	03018
206.	Finish improper	Visual	03019

4.3.2.4 Clip, safety (See Dwg. DXP-113789 covering a detail of Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Width of slot (for exposed length) max	Gage	04001
Major	**AQL 0.65 Percent		
101.	Thickness (prior to forming)	Gage	04002
102.	Width of leaf, min	Gage	04003
103.	Crack	Visual	04004
Major	**AQL 0.40 Percent (See Para 4.3.3.2)		
104.	Outside diameter	Gage	04005
105.	Total length	Gage	04006

**Combined AQL for Major Defects 101 through 105 (See Para 4.3.2) is as follows: AQL 1.00 Percent.

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Categories	Defects	Method of Inspection	Code No.
Minor	AQL 2.50 Percent (See Para 4.3.3.2)		
201.	Height to leaf, max	Gage	04007
202.	Width of tab, max	Gage	04008
203.	Length to full slot (from tab end) max	Gage	04009
204.	Height of tab	Gage	04010
205.	Length to notch (tab end)	Gage	04011
206.	Length of tab for indicated min depth, max	Gage	04012
207.	Evidence of poor workmanship (See 3.5)	Visual	04013

4.3.2.5 Pin, lower (See Dwg. Bp-119560 covering a detail of Dwg. DP-119556).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Profile of point	Gage	05001
102.	Length from shoulder to tip	Gage	05002
103.	Concentricity of tip with body	Gage	05003
104.	Diameter of body, min	Gage	05004
105.	Diameter of small end, min	Gage	05005
Minor	AQL 1.00 Percent		
201.	Radius improper	Visual	05006
202.	Evidence of poor workmanship (See 3.5)	Visual	05007

4.3.2.6 Spring, slide (See Dwg. CXP-113763 covering a detail of Dwg. FP-119554).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Solid height, max	Gage	06001
102.	Outside diameter, max	Gage	06002
103.	Inside diameter, min	Gage	06003
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	06004

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4.3.2.7 Spring, yoke, main (See Dwg. CP-119558 covering a detail of Dwg. DP-119556).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent (See Para 4.3.3.2)		
101.	Diameter of hole, max	Gage	07001
Minor	**AQL 1.00 Percent		
201.	Thickness	Gage	07002
202.	Evidence of poor workmanship (See 3.5)	Visual	07003
Minor	**AQL 2.50 Percent (See Para 4.3.3.2)		
203.	Width, min	Gage	07004
204.	Diameter, min	Gage	07005
205.	Symmetry of width with hole	Gage	07006
206.	Symmetry of distance across ends with hole	Gage	07007
207.	Concentricity of diameter with hole	Gage	07008
208.	Total length, max (after flattening)	Gage	07009

**Combined AQL for Minor Defects 201 through 208 (see paragraph 4.3.2) is as follows: AQL 4.00 Percent.

4.3.2.8 Spring, yoke, secondary (See Dwg. CP-119559 covering a detail of Dwg. DP-119556).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent (See para 4.3.3.2)		
101.	Diameter of hole, max	Gage	08001
Minor	**AQL 1.00 Percent		
201.	Thickness	Gage	08002
202.	Evidence of poor workmanship (See 3.5)	Visual	08003

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Minor	**AQL 2.50 Percent (See Para 4.3.3.2)		
203.	Width, min	Gage	08004
204.	Diameter	Gage	08005
205.	Symmetry of width with hole	Gage	08006
206.	Symmetry of distance across ends with hole	Gage	08007
207.	Concentricity of diameter with hole	Gage	08008

**Combined AQL for Minor Defects 201 through 207 (See para 4.3.2) is as follows: AQL 2.50 Percent.

4.3.2.9 Yoke (See Dwg. DP-119557) covering a detail of Dwg. DP-119556).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Diameter of lower pin hole, max	Gage	09001
102.	Height to top of slot	Gage	09002
Minor	AQL 1.50 Percent		
201.	Width of slot, max	Gage	09003
202.	Total height, min	Gage	09004
203.	Evidence of poor workmanship (See 3.5)	Visual	09005

4.3.2.10 Spring, outer (See Dwg. FP-119554 covering a detail of Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Length from face to end of safety clip bend, max	Gage	10001
Major	**AQL 0.65 Percent		
101.	Flatness of face	Gage	10002
102.	Location of hinge pin hole	Gage	10003
103.	Location of safety clip bend	Gage	10004
104.	Thickness	Gage	10005
Major	**AQL 0.40 Percent (See Para 4.3.3.2)		
105.	Diameter of hinge pin hole	Gage	10006

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Minor	**AQL 1.50 Percent		
201.	Angle of side improper, left side	Gage	10007
202.	Angle of side improper, right side	Gage	10008
203.	Evidence of poor workmanship (See 3.5)	Visual	10009
Minor	**AQL 0.65 Percent (See Para. 4.3.3.2)		
204.	Distance between sides for datum length (Note 13)	Gage	10010

**Combined AQL for Major Defects 101 through 105 (See paragraph 4.3.2) is as follows: AQL 1.00 Percent, and combined AQL for Minor Defects 201 through 204 (See paragraph 4.3.2) is as follows: AQL 1.50 Percent.

4.3.2.11 Housing, nose (See Dwg. FP-119548 covering a detail of Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	*AQL 0.65 Percent (See 4.3.3.4)		
101.	Profile at bottom of large cavity	Gage	11001
102.	Contour of propellant cavity counterbore	Gage	11002
103.	Depth from top to bottom of spherical radius	Gage	11003
104.	Distance from top to bottom of mouth angle at inside diameter (when applicable)	Gage	11004
Major	AQL 2.50 Percent		
105.	Symmetry outside flats with datums	Gage	11005
106.	Symmetry of large cavity with external datum widths	Gage	11006
107.	Total height	Gage	11007
108.	Length at top	Gage	11008
109.	Profile of top	Gage	11009
110.	Distance from top to first datum	Gage	11010
111.	Distance from top to second datum	Gage	11011
112.	Location of drive stud holes	Gage	11012
113.	Distance from top of housing to top of yoke spring slot, min	Gage	11013
114.	Diameter of propellant cavity flash hole	Gage	11014
115.	Diameter of propellant cavity	Gage	11015
116.	Concentricity of propellant cavity diameter with diameter of large cavity	Gage	11016

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117.	Diameter of large cavity	Gage	11017
118.	Total width	Gage	11018
119.	Depth of washer groove	Gage	11019
120.	Width of washer groove, min	Gage	11020
121.	Distance from top to base of propellant cavity	Gage	11021
122.	Distance from top to washer groove	Gage	11022
123.	Symmetry of large cavity with outside width	Gage	11023
124.	Diameter of slide cavity, max	Gage	11024
125.	Location of slide cavity	Gage	11025
126.	Depth of slide cavity, max	Gage	11026
127.	Symmetry of propellant cavity with datum widths	Gage	11027
128.	Symmetry of propellant cavity with outside widths	Gage	11028
129.	Periphery, max (See Note 9 on Dwg.)	Gage	11029
130.	Height of indicated area (See Note 13 on Drawing)	Gage	11030
131.	Diameter of drive stud hole, max (Spring end)	Gage	11031
132.	Diameter of drive stud hole, max (plug-rest end)	Gage	11032
133.	Depth of one yoke spring slot (spring end)	Gage	11033
134.	Depth of other yoke spring slot (plug-rest end)	Gage	11034
135.	Location of hinge pin holes	Gage	11035
136.	Radius missing from corner of diaphragm cavity	Visual	11037
137.	Burr in flash hole or slide cavity	Visual	11038

Minor — *AQL 1.00 Percent (See 4.3.3.4)

201.	Diameter of propellant cavity counterbore	Gage	11039
202.	Concentricity of propellant cavity counterbore with diameter of large cavity	Gage	11040

Minor — AQL 4.00 Percent

203.	Concentricity of propellant cavity flash hole with diameter of large cavity	Gage	11041
204.	Diameter of slide cavity vent hole, max	Gage	11042

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205.	Concentricity of slide cavity hole diameter with slide cavity diameter	Gage	11043
206.	Depth of slide cavity counterbore	Gage	11044
207.	Width of yoke slot, max	Gage	11045
208.	Radius at top of yoke slot, min	Gage	11046
209.	Total width, max (indicated area)	Gage	11047
210.	Evidence of poor workmanship (See 3.5)	Visual	11048
211.	Side cuts (4) on top of housing near hinge pin holes missing	Visual	11049

*Combined AQL for Major defects 101 through 137 is as follows: AQL 2.50 Percent.
Combined AQL for Minor defects 201 through 211 is as follows: AQL 4.00 Percent (See 4.3.2)

4.3.2.12 Yoke spring assembly (See Dwg. DP-119556 covering a detail of Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Symmetry of springs with slotted ends after application of torque	Test	12001
102.	Location of lower pin	Gage	12002
103.	Lower pin head above flush	Gage	12003
104.	Location of yoke springs (2)	Gage	12004
105.	Secondary yoke spring missing	Visual	12005
106.	Lower pin point damaged	Visual	12006
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	12007

4.3.2.13 Slide assembly (See Dwg. CP-119555 covering a detail of Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Primer or stake above flush	Gage	13001
102.	Primer inverted	Visual	13002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	13003

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4.3.2.14 Housing assembly (prior to assembly of plug-restricter assembly)(See Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Length from housing to top of hinge pin, max	Gage	14001
102.	Diaphragm missing	Visual	14002
Minor:	None defined		

4.3.2.15 Housing assembly (prior to assembly of yoke spring assembly)(See Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Slide in armed position	Visual	15001
Major	AQL 0.65 Percent		
101.	Depth to lower sleeve, max	Gage	15002
102.	Four tabs fail to engage sides of housing and lock outer spring when it is in fully open position	Manual	15003
103.	Slide reversed	Visual	15004
104.	Restricter missing from plug-restricter assembly	Visual	15005
Minor:	None defined		

4.3.2.16 Housing assembly (prior to assembly of safety clip and while slider in unarmed position)(See Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Head of drive stud above flush	Gage	16001
102.	Yoke spring assembly does not move freely	Manual	16002
Minor:	None defined		

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4.3.2.17 Housing assembly (after assembly of safety clip) (See Dwg. DP-119552).

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Depth to lower pin, max	Gage	17001
2.	Distance from outer surface of outer spring to spring end of slide assembly, max	Gage	17002
3.	Hinge pin (4) (a) missing (b) loose	Visual-Manual	17003
4.	Safety clip (a) missing (b) reversed, i.e, clip bend not on same side as red dot on housing and slide vent hole	Visual	17004
5.	Lower pin (a) too short (b) missing (either is indicated by primer being visible in bottom of large cavity)	Visual	17005
Major	AQL 0.65 Percent		
101.	Depth from side to safety clip, max	Gage	17006
102.	Outer springs extend above crown surface of housing	Gage	17007
103.	Red dot missing or on wrong side of housing	Visual	17008
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.5)	Visual	17009

4.3.3. Testing.

4.3.3.1 Dimensional control of cast parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least three (3) parts (as cast) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

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All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

4.3.3.2 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, with the approval of the Contracting Officer, a sample of at least ten (10) parts may be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.3.3.3 Check test for deterioration of primers (See Purchase Specification AEI-93) Major Defect (Code No. 18001).-If the total time between original acceptance of any lot and the assembly of that lot into the slide exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in Purchase Specification AEI-93 immediately before the primer lot is assembled into the slides. This test shall be performed by the contractor on primers selected by the Government Inspector at the facility assembling the primers into the slides. (6.4).

4.3.3.4 Tool control inspection covering the following housing characteristics: Diameter of propellant cavity counterbore, contour of propellant cavity counterbore, concentricity of propellant cavity counterbore with diameter of large cavity, profile at bottom of large cavity, depth from top to bottom of spherical radius and distance from top to bottom of mouth angle at inside diameter (when applicable) (for dwg number and code numbers, see 4.3.2.11).-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contract-

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ing Officer, the following inspection procedure may be used:

At the start of production, after production of each 2000 pieces from each tool, and whenever a change or modification of form tool is made during production, the required characteristic measurements shall be obtained from that tool. Determination of characteristics through the use of approved plastic or other impressions is permissible. The contractor's tolerance limits for control of characteristics shall be approved by the procuring activity. When a measurement falls outside the contractor's approved tolerance limits, fifty (50) nose housings from the latter portion of production from that tool, since the last acceptable measurement was taken, shall be measured for the required characteristics. The use of approved impressions is a permissible means of obtaining these characteristics for measurement. If any characteristic measurement taken in this manner falls outside the component drawing tolerance, the housings produced from that tool since the last satisfactory measurement was obtained shall be inspected one hundred percent for that characteristic by the contractor to remove all components in which one or more characteristics are outside drawing tolerances.

In addition, the required characteristic measurements shall be taken from five (5) nose housings from each eight hour's production or portion thereof from each production tool in use. The use of approved impressions is a permissible means of obtaining these characteristics for measurement. If any characteristic measurement falls outside of the drawing requirement, the nose housings produced since the last satisfactory measurement was obtained shall be inspected one hundred percent by the contractor to remove nose housings which do not comply with the applicable drawing requirement.

4.3.3.5 Mechanical properties of nose housing, (after heat treatment and prior to machining) Major defect (See Dwg. FP-119548).

4.3.3.5.1 Heat treatment lot (forgings) Defect Code 19001.- Three (3) nose housings shall be randomly selected from each heat treatment lot after completion of the heat treatment process, and the three (3) specimens prepared therefrom, shall be subjected to this test as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5.2 Continuous heat treatment lot (forgings) Defect Code 19002.-Three (3) nose housings shall be randomly selected from each eight hour's production, or portion thereof, after completion of the heat treatment process, and the three (3) specimens prepared therefrom shall be subjected to this test as specified in 4.4.1 using equipment in accordance with 4.3.4.

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4.3.3.5.3 Heat treatment lot (bar stock) Defect Code 19003.-Two (2) specimens shall be selected from each of three (3) bars, randomly selected from each heat treatment lot. The specimens shall be taken from one end and center of each of the selected bars. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4.

4.3.3.5.4 Rejection.-If any specimen fails to comply with the mechanical properties requirements specified on the applicable drawing, the heat treatment quantity represented by the failing specimen shall be rejected.

4.3.3.6 Hardness of hinge pin and outer spring, (See Dwg. CXP-113760 and FP-119554, resp.) Major defect (Code No. 20001.-Ten (10) of each part shall be selected from each heat treatment batch (See NOTE) and subjected to this test. If any part fails to comply with the requirement specified on the applicable drawing, the batch represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4.

NOTE 1-A heat-treatment batch shall be defined as parts that have been heat treated at the same time in the same furnace and quench bath for each of all phases of the entire heat treatment process.

4.3.3.7 Deflection.-Load test of leaf springs of outer spring (See Dwg. FP-119554) Major defect (Code No. 21001.-Eighty (80) outer springs shall be selected from each lot and subjected to this test. If three (3) or more parts fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. This test shall be prepared as specified in 4.4.3 using equipment in accordance with 4.3.4.

4.3.3.8 Spring test of side tabs of outer spring (See Dwg. FP-119554) Major defect (Code No. 22001).-After completion of test specified in 4.3.3.7, twenty-five (25) outer springs shall be subjected to this test. If any part fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.4 using equipment in accordance with 4.3.4.

4.3.3.9 Leakage test of diaphragm (See Dwg. CXP-113757) Major defect (Code No. 23001).-One hundred and twenty-five (125) diaphragms shall be selected from each lot and subjected to this test. If two (2) or more diaphragms fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.5 using equipment in accordance with 4.3.4.

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4.3.3.10 Deformation test of nose housing (See Dwg. FP-119548) Major defect (Code No. 24001).-Fifteen (15) nose housings shall be selected from each lot and subjected to this test. If any nose housing fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.6 using equipment in accordance with 4.3.4.

4.3.3.11 Direction of grain (Major defect) Code No. 25001.

a. Main yoke spring (See Dwg. CP-119558)
b. Secondary yoke spring (See Dwg. CP-119559)
c. Safety clip (See Dwg. DXP-113789)

Twenty-five (25) each of main yoke spring, secondary yoke spring and safety clip shall be selected from each lot and subjected to this test. If any part fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.7 using equipment in accordance with 4.3.4.

4.3.3.12 Pull load test of safety clip (See Dwg. DXP-113789) Major defect (Code No. 26001).-Twenty-five (25) safety clips shall be selected from each lot and subjected to this test. If any safety clip fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.8 using equipment in accordance with 4.3.4.

4.3.3.13 Load test of yoke (See Dwg. DP-119557) Major defect (Code No. 27001).-Twenty-five (25) yokes shall be selected from each lot and subjected to this test. If any yoke fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. This test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4.

4.3.3.14 Load tests of slide spring (after temperature conditioning) (See Dwg. CXP-113763) Major defect (Code No. 28001).-Fifty (50) slide springs shall be selected from each lot and subjected to these tests. If any slide spring fails to comply with the requirements specified on the applicable drawing, the lot shall be rejected. The tests shall be performed as specified in 4.4.10 using equipment in accordance with 4.3.4.

4.3.3.15 Salt spray test of hinge pins (See Dwg. CXP-113760) Major defect (Code No. 29001).

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4.3.3.15.1 Batch-type process.-When the parts are produced in a batch-type process, twenty-five (25) of each part shall be selected from each batch and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the batch represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.11.

4.3.3.15.2 Continuous-type process.-When the parts are produced in a continuous-type process, five (5) of each part shall be selected from each four (4) hours of production, or portion thereof, and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the quantity represented by the failing part shall be rejected. The test shall be performed as specified in 4.4.11.

4.3.3.15.3 Lot plan.-Twenty-five (25) of each part shall be selected from each lot and subjected to the salt spray test as specified on the applicable drawing. If any part fails to comply with the specified requirement, the lot represented by the failing sample shall be rejected. The test shall be performed as specified in 4.4.11.

4.3.3.16 Shear strength test of lower pin rod or coil stock (See Dwg. BP-119560) Major defect, Code No. 30001.-In rod form, two (2) specimens shall be selected from each rod of ten (10) rods randomly selected from each heat treatment lot. The specimen shall consist of one (1) taken from the middle and one (1) taken from either end. In coil form, two (2) specimens shall be taken from each end of each coil and two (2) specimens taken at the approximate center of each coil with a six (6) inch minimum distance between specimens. If any specimen fails to comply with the requirement specified on the applicable drawing, the heat treatment quantity represented by the failing specimen shall be rejected. The test shall be performed as specified in 4.4.12 using equipment in accordance with 4.3.4.

4.3.3.17 Push test of yoke springs and lower pin of yoke sprin assembly (See Dwg. DP-119556) Major defect (Code No. 31001).-Twenty-five (25) yoke spring assemblies shall be selected from each lot for this test. If any lower pin fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.13 using equipment in accordance with 4.3.4.

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4.3.3.18 Deflection.-Load test of yoke springs of yoke spring assembly (See Dwg. DP-119556) Major defect (Code No. 32001).-One hundred and twenty-five (125) yoke spring assemblies shall be selected from each lot and subjected to this test. If two (2) or more springs fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.14 using equipment in accordance with 4.3.4.

4.3.3.19 Push-out test of restricter of plug-restricter assembly (See Dwg. BXP-113764) Major defect (Code No. 33001).-One hundred and twenty-five (125) plug-restricter assemblies shall be selected from each lot and subjected to this test. If two (2) or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.15 using equipment in accordance with 4.3.4.

4.3.3.20 Air flow test of plug-restricter assembly (See Dwg. BXP-113764) Major defect (Code No. 34001).-One-hundred and twenty-five (125) plug-restricter assemblies shall be selected from each lot and subjected to this test. If two (2) or more assemblies fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.16 using equipment in accordance with 4.3.4.

CAUTION: The assemblies in the following tests contain an explosive component. Safety precautions shall be observed.

4.3.3.21 Push test of primer of slide assembly (See Dwg. CP-119555) Major defect (Code No. 35001).-One-hundred and twenty-five (125) slide assemblies shall be selected from each lot and subjected to this test. If two (2) or more primers fail to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.17 using equipment in accordance with 4.3.4.

4.3.3.22 Freeness of pivoting of outer springs (See Dwg. DP-119552) Major defect (Code No. 36001).-Three hundred and fifteen (315) housing assemblies shall be selected from each lot and subjected to this test. If six (6) or more assemblies fail to pivot freely, the lot shall be rejected. The test shall be performed as specified in 4.4.18 using equipment in accordance with 4.3.4.

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4.3.3.23 Security of hinge pins and outer springs (See 3.3) Major defect (Code No. 37001).-Fifty (50) assemblies shall be selected from each lot and subjected to the drop test. If during the test, eleven (11) or more holes in outer springs, or two (2) or more hinge pins fail to meet the requirements of paragraph 3.3, the lot shall be rejected. The ten (10) failures allowed for holes in outer springs shall contain no more than a total of two (2) failures wherein either a section of metal between the hole and edge is torn out, or a hole in a corner of a spring has pulled free from the hinge pin. The test shall be performed as specified in 4.4.19 using equipment in accordance with 4.3.4.

4.3.3.24 Nose housing qualification test (after final heat treatment) (See Dwg. FP-119548) Major defect (Code No. 38001).-One-hundred and twenty (120) nose housings shall be selected from each lot and subjected to this test. The test shall be performed and test data evaluated in accordance with 4.4.20 using equipment in accordance with 4.3.4. If the resultant value does not exceed three and one-half ($3\frac{1}{2}$) inches, the lot shall be rejected and additional nose housings shall be selected and tested as specified in 4.3.3.24.1.

4.3.3.24.1 Simulated grenade catapult test (See 3.4 and Dwg. DP-119552) Major defect (Code No. 39001) (when applicable).-One-hundred and twenty-five (125) housing assemblies shall be selected from each lot (See Note in para. 4.3.1), prepared as specified in 6.5, and subjected to this test as specified in 4.4.21. Equipment shall be in accordance with 4.3.4. If three (3) or more housing assemblies fail to meet the specified requirements, the lot shall be rejected.

4.3.3.25 Delay arming time of housing assembly (See Dwg. DP-119552) Major defect (Code No. 40001).-Two-hundred (200) housing assemblies shall be selected from each lot and subjected to this test. The lot shall be rejected if two (2) or more assemblies exceed six seconds, or a total of eight (8) or more assemblies exceed the delay arming time requirements specified on the applicable drawing. The test shall be performed as specified in 4.4.22 using equipment in accordance with 4.3.4.

4.3.3.26 Push-out test of plug-restricter assembly (See Dwg. DP-119552) Major defect (Code No. 41001).-After completion of test specified in 4.3.3.25, twenty-five (25) housing assemblies shall be selected and subjected to this test. If any plug-restricter assembly fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.23 using equipment in accordance with 4.3.4.

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4.3.3.27 Push test of lower sleeve in housing assembly (See Dwg. DP-119552) Major defect (Code No. 42001).-After completion of test specified in 4.3.3.26, the twenty-five (25) housing assemblies shall be subjected to this test. If any lower sleeve fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected. The test shall be performed as specified in 4.4.24 using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number ET-119552, and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures.

4.4.1 Mechanical properties of nose housing (after heat treatment and prior to machining).-The test specimen shall be selected as indicated below and prepared and subjected to the tension tests to determine the mechanical properties specified on the applicable drawing. Test procedures shall be as specified in ASTM E8-66.

a. Forgings (See Figure 1).
b. Bar stock (Figure 9, Specimen 1 of ASTM E8-66. Shall be taken from the center of the area of the largest possible inscribed circle on the face of the nose housing).

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4.4.2 Hardness of hinge pin and outer spring.-These parts shall be subjected to the Rockwell hardness test using procedures specified in ASTM E18-67.

4.4.3 Deflection-load test of leaf springs of outer spring.-The outer spring shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied to the leaf spring. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.4 Spring test of side tabs of outer spring.-The outer spring shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied to the tabs. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.5 Leakage test of diaphragm.-The diaphragm shall be assembled in a fixture as shown on the Equipment List and a measured quantity of air applied to produce the air pressure specified on the applicable drawing. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.6 Deformation test of nose housing.-The nose housing shall be assembled in a fixture as shown on the Equipment List and the load specified on Dwg. FP-119548 applied to the nose housing. Reading for deformation shall be checked under load. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.7 Direction of grain.-The test shall be conducted on the specified parts using any method satisfactory to the Contracting Officer. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.8 Pull load test of safety clip.-The safety clip shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied. Observation shall be made for distortion in excess of tolerance. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.9 Load test of yoke.-The yoke shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied to the yoke. This is a non-destructive test. Parts so tested may be returned to the lot.

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4.4.10 Load tests of slide spring (after temperature conditioning).-The slide spring shall be assembled in a fixture and tested as specified on the Equipment List. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.11 Salt spray test of hinge pins.-These parts shall be subjected to the salt spray test in accordance with ASTM B 117-64 using a 5% atomized sodium chloride solution. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.12 Shear strength test of lower pin rod or coil stock.-The rod stock or coil stock specimen shall be assembled in a fixture as shown on the Equipment List and the load applied as specified on the applicable drawing. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.13 Push test of yoke springs and lower pin of yoke spring assembly.-The yoke spring assembly shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied to the lower pin. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.14 Deflection-load test of yoke springs of yoke spring assembly.-The yoke spring assembly shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied to the springs. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.15 Push-out test of restricter of plug-restricter assembly.-The plug-restricter assembly shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied to the restricter. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.16 Air flow test of plug-restricter assembly.-The plug-restricter assembly shall be assembled in a fixture as shown on the Equipment List and a measured quantity of air applied to produce the required air pressure specified on the applicable drawing. This is a non-destructive test. Parts so tested may be returned to the lot.

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4.4.17 Push test of primer of slide assembly.-The slide assembly shall be assembled in a fixture as shown on the Equipment List and the load specified on the applicable drawing applied to the primer. This is a non-destructive test. Parts so tested may be returned to the lot.

4.4.18 Freeness of pivoting of outer springs.-The housing assembly shall be mounted in a fixture as shown on the Equipment List and tested as follows:

> Assemblies having one or both sides failing the requirements of Para 4.3.3.22 shall be considered acceptable if a maximum load of 1.5 pounds causes the sticky side or sides to move to the proper position.

4.4.19 Security of hinge pins and outer springs.-The housing assembly shall be assembled in a fixture as shown on the Equipment List and the load specified in 3.3 applied to hinge pins and outer springs. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.20 Nose housing qualification test (after final heat treatment).-The nose housing shall be modified, assembled in test fixture, tested and data evaluated, as required by Operating instructions, Special Instructions, etc. specified on Equipment List EL-119548.

4.4.21 Simulated grenade catapult test (when applicable).-The housing assemblies shall be modified, loaded, then assembled in a test fixture, tested and data evaluated as specified by Operating Instructions, Special Instructions, etc. as specified on the Equipment List. (See 6.5 and 6.6).

4.4.22 Delay arming time of housing assembly.-The housing assembly shall be assembled in a fixture and tested as specified on the Equipment List. Observation shall be made that the slide moves into arming position in the time specified on the applicable drawing. If the lot is acceptable, the remaining eighty-five (85) housing assemblies that were subjected only to the arming time delay test, shall be disarmed in a fixture as specified in the Inspection Equipment List and returned to the lot. They shall be disarmed with a maximum pressure (50) p.s.i. Assemblies that do not pass the time delay, shall not be disarmed and returned to the lot.

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4.4.23 Push-out test of plug-restricter assembly.-The housing assembly shall be drilled out and the slide assembly removed. Assemble the housing assembly in a fixture as specified on the Equipment List and apply the load specified on the applicable drawing to the plug-restricter assembly. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.24 Push test of lower sleeve in housing assembly.-The housing assembly shall be assembled in a fixture as specified on the Equipment List and the load specified on the applicable drawing applied to the lower sleeve. This is a destructive test. Parts so tested shall not be returned to the lot.

4.4.25 Initial qualifying test for housing assembly pack.-The fiber box or wooden box, with housing assemblies, shall be subjected to the drop tests as specified in Code of Federal Regulations, Title 49, Parts 171-179, Section 78.214-20. Assemblies and components shall be examined to determine conformance to requirements specified in 3.6.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging-

5.1.1 Level C.

5.1.1.1 Housing assembly (Dwg. DP-119552).-One-hundred and eighty (180) housing assemblies shall be packed in 275 lb. min test Type 1, Class 1, "B" flute, RSC Style fiber box, Specification PPP-B-636 on formed trays of high impact polystyrene or other approved material. A minimum of 30 and maximum of 60 assemblies shall be packed per tray. The tray shall contain individual triangular-shaped compartments for the housing assembly, and shall provide a recess to prevent any pressure on the protruding safety clip. A partition of the same material as the fiber box shall be used between each layer of trays and as required for tight packing. Maximum moisture content of fiber box and partitions shall be 9% prior to packaging.

All corners of sealed fiber box shall be blunted. The fiber box shall be inserted in Type 1, Class E, MIL-B-117 bag. Prior to sealing bag, exhaust all excess air within bag. Each shipping lot may contain one partial tray and box.

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5.1.2 Additional hardware.-See 5.2.1.2.

5.2 Packing.

5.2.1 Level C.

5.2.1.1 Housing assembly (Dwg. DP-119552) (See 3.6).-The housing assemblies, preserved and packaged as specified in 5.1.1.1, shall be inserted in Type 1, Class 1, Grade C, MIL-B-2427 wooden box. Commercial chipboard filler material shall be added as required for tight packing. The box shall be sealed with two metal straps 5/8 x .02 inches. As an alternative to the wooden box, an equivalent approved fiber box complying with Section 78.214, Spec 23F, of Code of Federal Regulations, Title 49, Parts 171-179 may be used.

5.2.1.2 Additional hardware.-Additional hardware shall be packed in accordance with MIL-STD-1169.

5.3 Marking.

5.3.1 Housing assembly.

5.3.1.1 Fiber box (inner) and bag.-Marking shall be in accordance with MIL-STD-129 and Code of Federal Regulations, Title 49, Parts 171-179 and the following:

"180 Housing assembly DP-119552
Date Packed Lot"

5.3.1.2 Wooden box.-Wooden boxes containing the housing assemblies shall be marked in accordance with Dwg. C8796522. Code of Federal Regulations marking to be, "Percussion Fuzes - Handle Carefully".

5.3.1.3 Alternate fiber box.-Marking shall be in accordance with MIL-STD-129. In addition, the cover shall be marked, "Percussion Fuzes - Handle Carefully".

5.3.2 Additional hardware.-Marking shall be as specified in MIL-STD-129 and Code of Federal Regulations, Title 49, Parts 171-179.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

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6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings of Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the primer assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the primer assemblies, or to the contractor assembling the primer assemblies into the sliders; and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer assembly lot was under the jurisdiction of either contractor.

6.5 Preparation of item for functional testing.-The sample housing assembly (DP-119552) shall be modified as follows: A standard one-sixteenth (1/16) inch hole shall be drilled into and at the following location: Center the 1/16 inch drill one-thirty second (1/32) inch above the seam (between the plug and restricter.

Caution: Care shall be taken so as to prevent chips from entering the hole by use of compressed air, or other suitable means, during drilling.

The item shall then be loaded in accordance with the normal loading operations and tested as specified on the applicable drawing of the Equipment List.

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6.6 Submission of test data.-In addition to the normal distribution of records, when the Housing Assembly is procured by the U. S. Army Munitions Command, one (1) copy of all Catapult Test data and ammunition data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

6.7 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

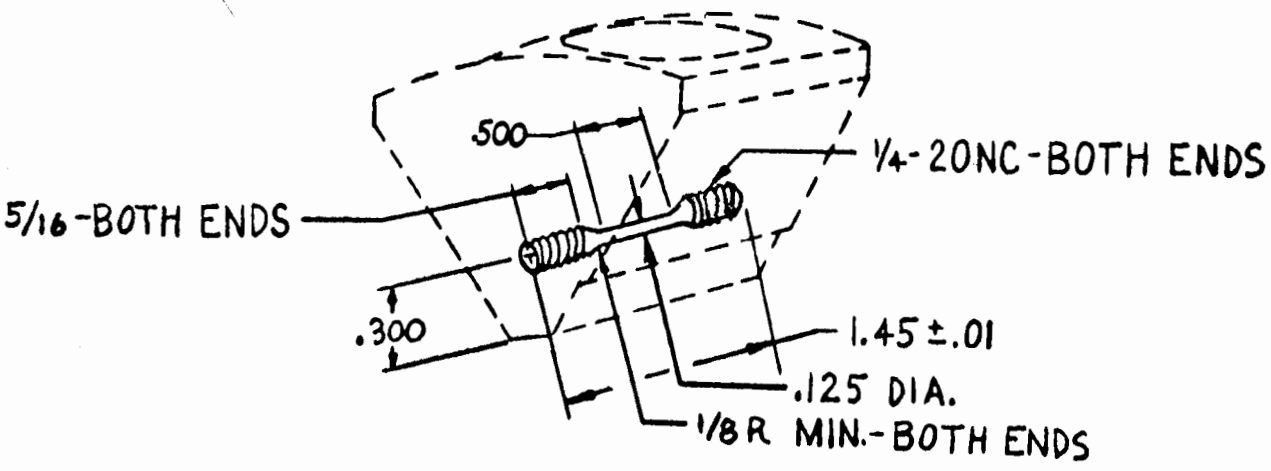
6.8 Supersession data.-This specification includes the requirements of Purchase Specification AEI-162B, dated 6 August 1966.

Custodian:
Army-MU

Preparing Activity:
Army-MU

Project Number: 1330-A208

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SPECIMEN FOR TENSION TEST
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FIGURE 1

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